

HCLS AI Agent Suite

Governed Agentic AI for Life Sciences — Built on AWS

Eight reference agents on one 21 CFR Part 11 / GxP-defensible platform — the agent isn't the product, the governance is.

EVERYONE'S MOVING. FEW ARE GOVERNED.

In a workflow where a hallucinated number is a data-integrity defect, governance is the gap this platform closes.

\$60–110B

/yr potential gen-AI value for pharma & medical products

[industry-research — McKinsey 2023]

~5%

of commercial life-sci firms have turned gen AI into real value

[industry-research — McKinsey 2024]

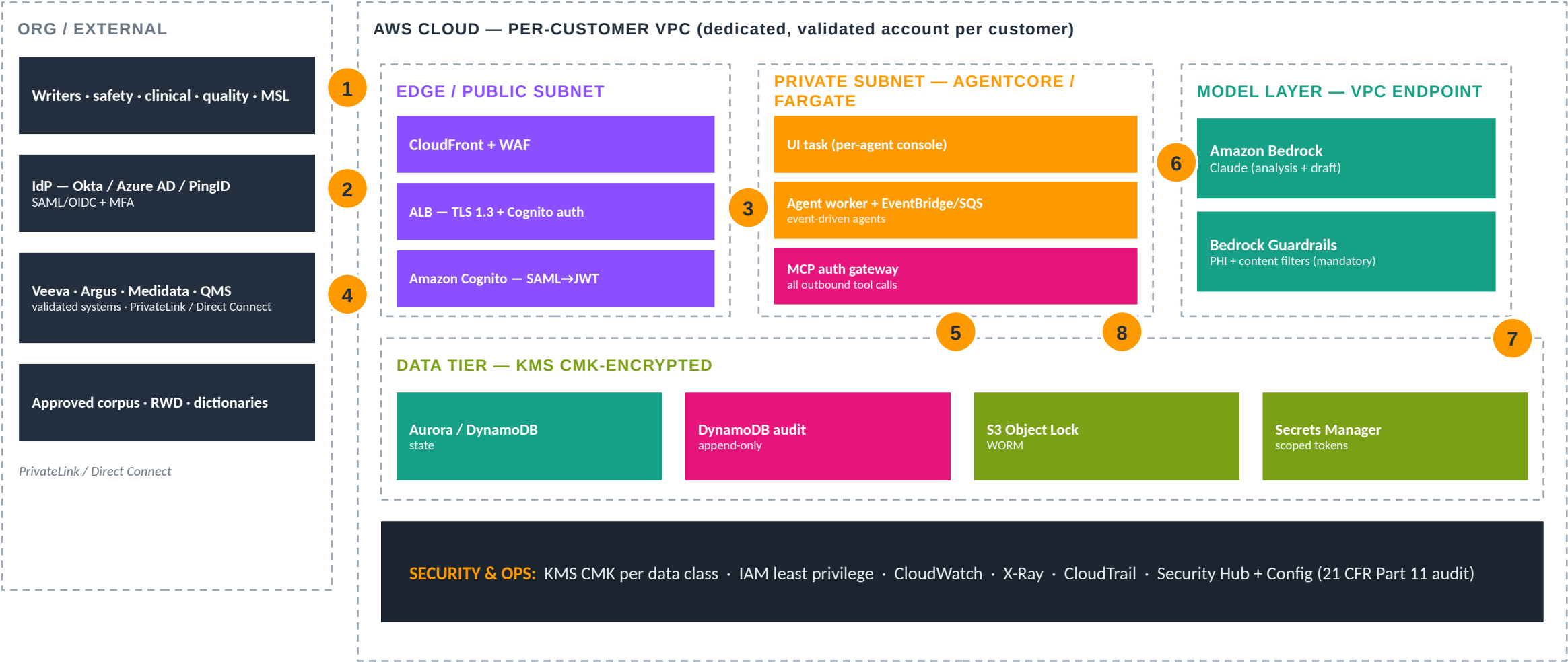
Jan 2025

FDA draft guidance on AI in regulatory decision-making (7-step credibility)

[gov/peer-reviewed — FDA]

The takeaway: this platform is the governance layer that lets a sponsor adopt AI as fast as the market — and defend it to an FDA/EMA inspector.

Shared AWS Architecture & Traffic Flow



1 sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT — no credentials in AWS) 3 authenticated session 4 systems-of-record read over private connectivity 5 queue/scale for event-driven agents 6 model calls stay in-VPC via Bedrock + Guardrails endpoint (PHI masked first) 7 every action persisted to append-only audit (21 CFR Part 11) 8 tools reachable only through the governed MCP gateway

The 8-Agent Portfolio — Land-first (document- & case-centric)

01 Regulatory Writing & Intelligence

draft and assemble submission content from grounded evidence — a qualified human signs off

Headline: \$2.6B — average capitalized cost per approved drug — the value behind every submission

Cost of doing nothing: ~\$60M / month

02 Pharmacovigilance — ICSR Case Intake

triage, de-duplicate, MedDRA-code, and draft E2B narratives — a safety physician validates the case

Headline: ~28M — cumulative FAERS reports by end-2023; quarterly volume ~3× since 2007

Cost of doing nothing: ~\$2.0M / yr

03 Clinical Trial Ops & TMF

surface TMF gaps and EDC query backlog continuously — a CRA approves every disposition

Headline: ~\$800K / day — lost/delayed sales per day of trial delay (+ ~\$40K/day direct cost)

Cost of doing nothing: ~\$25.7M / slip

08 Medical Affairs / MSL Copilot

draft on-label, evidence-grounded HCP responses — MLR review approves before anything ships

Headline: weeks→months — MLR review is the single biggest content-release bottleneck

Cost of doing nothing: Billions in FCA risk

The 8-Agent Portfolio — Higher-governance (sequence after)

04 Site Selection & Patient Matching

rank sites on real performance and size cohorts on RWD — humans select sites and confirm eligibility

Headline: ~80% — of trials miss their original enrollment timeline

Cost of doing nothing: ~\$24M / launch

05 Quality / CAPA & Complaints

draft root-cause and CAPA, score complaints — a quality owner approves classification and closure

Headline: #1 cited — CAPA (21 CFR 820.100) is consistently the most-cited FDA device 483 clause

Cost of doing nothing: \$10M-\$100M / recall

06 Clinical Protocol Design & Feasibility

draft protocol sections grounded in guidance and RWD — clinical leads own the design

Headline: ~57% — of protocols incur ≥ 1 substantial amendment; ~45% avoidable

Cost of doing nothing: ~\$535K / amendment

07 Real-World Evidence / HEOR

build cohorts and translate code so analysts analyze — an epidemiologist approves the study

Headline: ~45% — of analyst time goes to data prep, not analysis (measured)

Cost of doing nothing: ~\$1.3M / yr / team

The Governance & Compliance Spine

Every agent inherits the same controls — mapped to the regulations life-sciences buyers are inspected against.

Regulation / framework	Mapped control on the platform
21 CFR Part 11 (records / e-sig)	Append-only audit; bound reviewer identity on every gate; system validation posture; KMS CMK encryption
GxP / ALCOA+ (data integrity)	Grounding verification; every figure traceable to source; prompt version pinning; tamper-evident lineage
GVP / ICH E2B(R3) (safety)	PHI masked before model calls; safety-physician gate on causality & reportability (Agent 02)
ICH E6(R3) GCP (trials)	Continuous TMF completeness; CRA gate on dispositions; inspection-ready audit (Agent 03)
HIPAA / de-identification	Safe Harbor / Expert Determination enforced at the gateway for any RWD use (Agents 04/07)
FDA AI guidance (Jan 2025) + NIST AI RMF	Risk-based credibility framing; Guardrails, explainable grounded output, framework-enforced HITL

The Maturity Ladder

Climb at the sponsor's own pace — each rung reuses the same governed control plane.

1 · Documented

Architecture, workflow, and compliance design written and reviewed. Useful for discovery and architecture review.

2 · Demonstrated

Runs end-to-end in demo mode (no API key, deterministic fixtures). Internal demos and customer workshops.

3 · Deployable

CloudFormation + container contracts + CI pass; customer AWS account + Bedrock. Suitable for a pilot.

4 · Production-ready

CSV/CSA complete, IdP integrated, connectors live, pen-test passed. The engagement milestone.

The suite sits at Demonstrated + Deployable-by-design — 452 tests pass with no API key; production-readiness is the engagement.

Deployment — One Platform, Per-Customer Isolation

1 Foundation

Per-customer VPC + dedicated, validated AWS account; KMS CMK per data class; network to systems-of-record (PrivateLink / Direct Connect).

2 Identity

Cognito SAML→JWT federation with role scoping (writer / safety / CRA / quality / MSL / reviewer). No credentials stored in AWS.

3 Edge

CloudFront + WAF; ALB TLS 1.3; Cognito auth at the edge.

4 Agent + tools

Deploy the AgentCore/Fargate agent; grant tools via the deny-by-default MCP gateway with short-lived scoped tokens.

5 Connectors + data

Wire Veeva / Argus / Medidata / QMS connectors and Secrets Manager; index approved corpus; add per-agent blocks (EventBridge, Step Functions HITL).

6 Governance + go-live

Bedrock Guardrails (mandatory); S3 Object Lock + DynamoDB append-only audit; CloudTrail / Security Hub / Config; smoke + HITL test.

Reference: [docs/DEPLOY-QUICKSTART.md](#) · [docs/DEPLOYMENT-HANDBOOK.md](#) · [aws-native-reference/<agent>/DEPLOY.md](#)

Land-and-Expand

Prove the governed control plane with document- and case-centric agents, then expand to higher-governance workloads on the same foundation.

LAND

01 Regulatory Writing · 02 Pharmacovigilance (live path) · 03 Trial Ops/TMF · 08 Medical Affairs

Document- and case-centric, measurable. Proves identity, gateway, PHI masking, audit, and Guardrails in production.

EXPAND

05 Quality/CAPA · 06 Protocol Design

Add reportability and design-ownership gates on the now-proven control plane.

SCALE

04 Site/Patient Matching · 07 RWE/HEOR

Highest-governance: de-identified RWD, enforced at the gateway, with auditable lineage.

The Suite-Level Cost of Inaction

Per-agent cost of doing nothing, at a reference sponsor — substitute your own volumes.

01 Reg Writing

~\$60M / month

02 Pharmacovigilance

~\$2.0M / yr

03 Trial Ops / TMF

~\$25.7M / slip

04 Site / Patient

~\$24M / launch

05 Quality / CAPA

\$10M–\$100M / recall

06 Protocol Design

~\$535K / amendment

07 RWE / HEOR

~\$1.3M / yr / team

08 Medical Affairs

Billions in FCA risk

Several are modeled at a reference baseline (shown as such); 01/03/05/06 lean on direct published figures. None are guaranteed — they substitute the customer's actuals.

The Takeaway

The agent isn't the product. The governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.

Eight reference agents · one control plane · land-first with 01 / 02 / 03 / 08 · expand at your own pace.